Earlier Problem IGNORE:

```
Q = [2 2; 2 6];
```

```
Q = 2×2
     2     2
     2     6
```

```
C = [2; 10];
```

```
C = 2×1
     2
    10
```

```
pinv(Q)*Q;
```

```
ans = 2×2
    1.0000         0
    0.0000    1.0000
```

```
x_min = -pinv(Q) * C;
```

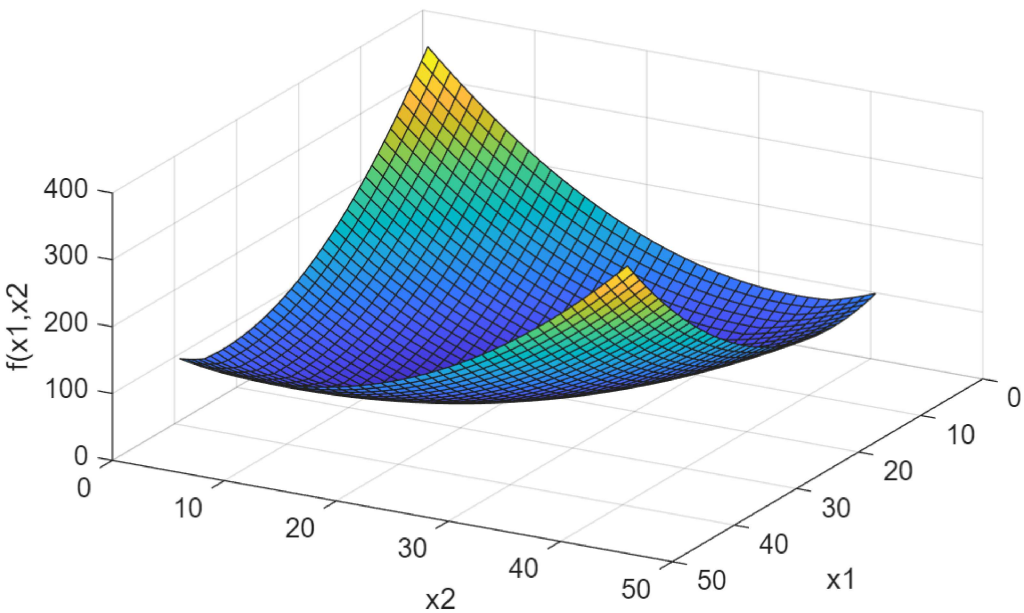
```
x_min = 2×1
    1.0000
   -2.0000
```

```
rank(Q);
```

```
ans =
2
```

```
%Starting here, you can define a function f(x1,x2) and plot it. If you want
%a heatmap, change the "surf(f)" command to "map(f)"

x1 = -10:.5:10;
x2 = -10:.5:10;

clear min1
for i = 1:size(x1,2)
    for ii = 1:size(x2,2)
    f(i,ii) = x1(i)^2 + x1(i)*x2(ii) +x2(ii)^2 + 3*abs(x1(i) + x2(ii) + 2);
        if exist("min","var") ~= 1
            min1 = [i, ii, f(i,ii)];
        end
        if f(i,ii) < min(3)
            min1 = [i, ii, f(i,ii)];
        end
    end
end
hold on
surf(f)
view([119 38])
xlabel("x1")
```

```
ylabel("x2")
zlabel("f(x1,x2")
grid("on")

plot3(min1(1), min1(2), min1(3));

hold off
```



HW2 Problem3, Jet Thrust:

```
figure
syms rho S C K W T T_diff V V_min T_min
res = 1;

T = 0.5*rho*S*V^2*C + (K*W^2)/ (0.5*rho*S*V^2)  %orig function
```

T =

$$\frac{C\,S\,V^2\,\rho}{2}+\frac{2\,K\,W^2}{S\,V^2\,\rho}$$

```
dT_dV = diff(T,V)   %first deriv
```

dT_dV =

$$C\,S\,V\,\rho-\frac{4\,K\,W^2}{S\,V^3\,\rho}$$

```
d2T_dV2 = diff(dT_dV,V)     %second deriv
```

d2T_dV2 =

$$C\,S\,\rho+\frac{12\,K\,W^2}{S\,V^4\,\rho}$$

```
V_mins = solve(dT_dV ==0,V)
```

V_mins =

$$\begin{pmatrix} -\sigma_1 \\ -\sigma_1\,\mathrm{i} \\ \sigma_1\,\mathrm{i} \\ \sigma_1 \end{pmatrix}$$

where

$$\sigma_1 = \left(\frac{4\,K\,W^2}{C\,S^2\,\rho^2}\right)^{1/4}$$

```
V_min = V_mins(4)
```

V_min =

$$\left(\frac{4\,K\,W^2}{C\,S^2\,\rho^2}\right)^{1/4}$$

```
T_min = simplify(subs(T,V,V_min))
```

T_min =

$$\frac{2\,K\,W^2}{S\,\rho\,\sqrt{\frac{K\,W^2}{C\,S^2\,\rho^2}}}$$

```
d2T_dV2_min = subs(d2T_dV2,V,V_min) %second deriv at min candidate V_min
```

d2T_dV2_min = $4\,C\,S\,\rho$

```
%eqns below are for numerical values of W, S, C, K, rho
T_num = vpa(subs(T,[W S C K rho], [73000 950 0.015 0.05 1.7553e-3]))
```

T_num =

$$\frac{319573502.20535101164464788299892}{V^2} + 0.012506512500000000018707951854324\,V^2$$

```
T_min_num = vpa(subs(T_min,[W S C K rho], [73000 950 0.015 0.05 1.7553e-3]))
```

T_min_num = $3998.3746697877126282358794144459$

```
V_min_num = vpa(subs(V_min,[W S C K rho], [73000 950 0.015 0.05 1.7553e-3]))
```

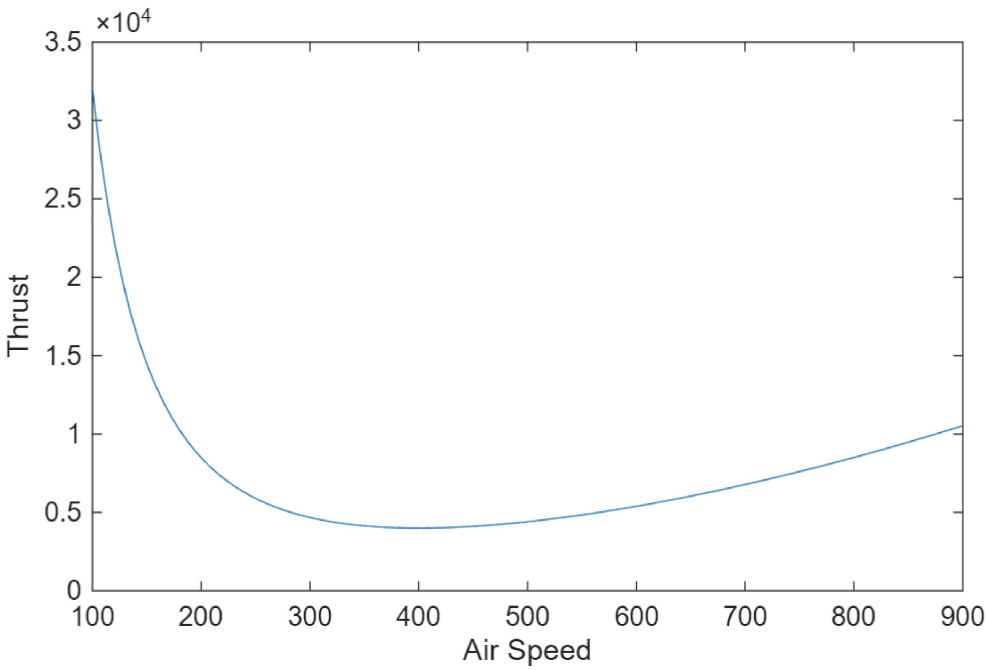
V_min_num = $399.81458709468884134237590657996$

```
V_num = 100:res:900;
```

```
T_num = double(subs(T_num, V, V0))
```

```
T_num = 1×801
10^4 ×
    3.2082    3.1455    3.0847    3.0256    2.9682    2.9124    2.8582    2.8056 ···
```

```
plot(V_num,T_num)
xlabel("Air Speed")
ylabel("Thrust")
```




```
[min_T, min_index] = min(T_num);
min_T
```

```
min_T =
3.9984e+03
```

```
minimizer_V = V_num(min_index)
```

```
minimizer_V =
400
```

```
minimizer_index = min_index;
```